

LAB No. 14

User and Group Management in Linux

Objective:

To learn how to **create, delete, and modify users and groups**, manage user passwords, assign permissions, and understand access control in Linux.

Learning Outcomes

After completing this lab, students will be able to:

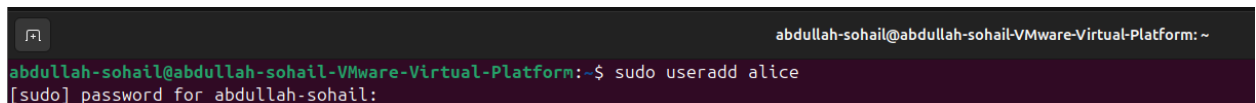
1. Create, modify, and delete user accounts and groups.
2. Set and manage user passwords.
3. Assign users to primary and secondary groups.
4. Understand and change file ownership and permissions.
5. Apply sudo privileges for administrative tasks.

Lab Tasks and Commands

Task 1: Creating New Users

1. Open a terminal and switch to the root user or use sudo.
2. Create a new user:

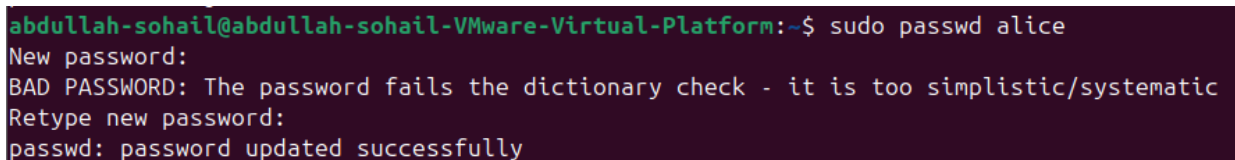
```
sudo useradd john
```



```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo useradd alice  
[sudo] password for abdullah-sohail:
```

3. Assign a password to the user:

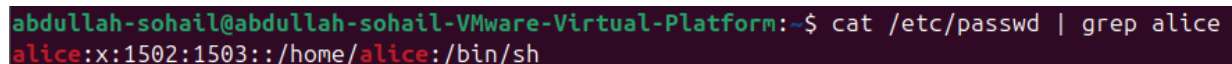
```
sudo passwd john
```



```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo passwd alice  
New password:  
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic  
Retype new password:  
passwd: password updated successfully
```

4. Verify that the user is added:

```
cat /etc/passwd | grep john
```



```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ cat /etc/passwd | grep alice  
alice:x:1502:1503:./home/alice:/bin/sh
```

5. Switch to the new user account:

```
su - john
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ su - alice
Password:
su: warning: cannot change directory to /home/alice: Permission denied
```

6. Check the home directory and shell assigned:

```
pwd
echo $SHELL
```

```
$ pwd
/home/alice_new
$ echo $SHELL
/bin/sh
```

Task 2: Modifying User Accounts

1. Change the login shell for a user:

```
sudo usermod -s /bin/sh john
```

```
$ sudo usermod -s /bin/sh alice
usermod: no changes
```

2. Change the user's home directory:

```
sudo usermod -d /home/john_new john
```

```
$ sudo usermod -d /home/alice_new alice
usermod: no changes
```

3. Then create and move the directory if needed:

```
sudo mkdir /home/john_new
sudo mv /home/john/* /home/john_new/
sudo chown -R john:john /home/john_new
```

```
$ sudo mkdir /home/alice_new
$ sudo mv /home/alice/* /home/alice_new/
$ sudo chown -R alice:alice /home/alice_new
```

4. Lock and unlock user accounts:

```
sudo usermod -L john # Lock
sudo usermod -U john # Unlock
```

```
$ sudo usermod -L alice
$ sudo usermod -U alice
```

5. Add user to additional groups:

```
sudo usermod -aG sudo john
```

```
$ sudo usermod -aG sudo alice
$ groups alice
alice : alice sudo
$ id alice
uid=1502(alice) gid=1503(alice) groups=1503(alice),27(sudo)
$
```

Task 3: Deleting Users

1. Delete a user (without removing home directory):

```
sudo userdel john
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo userdel alice
```

2. Delete a user **along with their home directory**:

```
sudo userdel -r john
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo userdel -r alice
userdel: user 'alice' does not exist
```

3. Verify removal:

```
grep john /etc/passwd
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ grep alice /etc/passwd
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

Task 4: Group Management

1. Create a new group:

```
sudo groupadd developers
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo groupadd developers
groupadd: group 'developers' already exists
```

2. View all groups:

```
cat /etc/group
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,abdullah-sohail
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
proxy:x:13:
kmem:x:15:
dialout:x:20:
```

3. Add a user to a group:

```
sudo usermod -aG developers john
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo useradd alice
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo usermod -aG developers alice
```

4. Change the primary group of a user:

```
sudo usermod -g developers john
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo usermod -g developers alice
```

5. Remove a group:

```
sudo groupdel developers
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo groupdel developers
groupdel: cannot remove the primary group of user 'alice'
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

Task 5: Understanding File Ownership and Permissions

1. Create a file:

```
touch testfile.txt
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ touch testfile.txt
```

2. Check ownership:

```
ls -l testfile.txt
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls -l testfile.txt
-rw-rw-r-- 1 abdullah-sohail abdullah-sohail 0 Apr 26 15:29 testfile.txt
```

3. Change ownership to a specific user and group:

```
sudo chown john:developers testfile.txt
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo chown alice:developers testfile.txt
```

4. Change file permissions:

```
chmod 740 testfile.txt
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ chmod 740 testfile.txt
```

```
chmod: changing permissions of 'testfile.txt':
```

5. `ls -l testfile.txt`

- Owner: Read, write, execute
- Group: Read only
- Others: No permission

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls -l testfile.txt
-rw-rw-r-- 1 alice developers 0 Apr 26 15:29 testfile.txt
```

6. Switch to the user john and test file access:

```
su - john
```

```
cat /home/yourusername/testfile.txt
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ su - alice
Password:
$ cat testfile.txt
```

Task 6: Granting Administrative Privileges

1. Open the sudoers file securely:

sudo visudo

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo visudo
```

```
GNU nano 7.2 /etc/sudoers.tmp
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"
adminuser ALL=(ALL) NOPASSWD: /bin/systemctl
# This fixes CVE-2005-4890 and possibly breaks some versions of kdesu
# (#1011624, https://bugs.kde.org/show_bug.cgi?id=452532)
Defaults        use_pty

# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"
```

2. Add the following line to allow a user administrative access:

john ALL=(ALL:ALL) ALL

```
# See sudoers(5) for more information on "@include" directives:

@include /etc/sudoers.d
alice ALL=(ALL:ALL) ALL
```

3. Save and exit, then test with:

su - john

sudo ls /root

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ su - alice
Password:
$ sudo ls /root
[sudo] password for alice:
acl_test myfile.txt snap testfile.txt test.txt
```